

OPERATING SYSTEMS

WEEK-1 Practical

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Section no:- 9

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Program:-

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<sys/wait.h>
int main(){
char cmd[10];
pid_t pid;
printf("Enter the command\n");
scanf("%s", cmd);
pid = fork();
if(pid == 0){
execlp(cmd,cmd,NULL);
printf("exec() sc failed");
exit(1);
}
else{
wait(NULL);
printf("parent pid:%d",getppid());
printf("child pid:%d",getpid());
}
return 0;
}
```

Output:-

```
Enter the command
uname
Linux
parent pid:329child pid:617dhanu@dhnpc:~$ |
```

79908

```
Enter the command
lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 356.9M  1 disk
sdb   8:16     0 159.4M  1 disk
sdc   8:32     0    2G    0 disk [SWAP]
sdd   8:48     0    1T    0 disk /mnt/wslg/distro

parent pid:329child pid:689dhanu@dhnpc:~$ |
```

```
top - 06:55:37 up 10 min, 1 user, load average: 0.06, 0.02, 0.00
Tasks: 28 total, 1 running, 27 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.6 total, 7176.1 free, 550.7 used, 209.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7233.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
88	root	20	0	35380	12356	9256	S	0.3	0.2	0:00.41	systemd-udevd
1	root	20	0	24124	15368	11568	S	0.0	0.2	0:00.60	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
8	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
48	root	19	-1	50404	16760	15472	S	0.0	0.2	0:00.16	systemd-journal
80	systemd+	20	0	22416	14436	11940	S	0.0	0.2	0:00.06	systemd-resolve
153	root	20	0	2888	1944	1820	S	0.0	0.0	0:00.06	chronyd-starter
154	root	20	0	4472	3120	2868	S	0.0	0.0	0:00.00	cron
155	message+	20	0	8792	5460	4780	S	0.0	0.1	0:00.05	dbus-daemon
165	root	20	0	44096	29892	14752	S	0.0	0.4	0:00.28	networkd-dispat
173	root	20	0	18436	9416	8168	S	0.0	0.1	0:00.08	systemd-logind
196	root	20	0	1792300	14364	11780	S	0.0	0.2	0:00.14	wsl-pro-service
214	syslog	20	0	220548	5672	4364	S	0.0	0.1	0:00.07	rsyslogd
254	root	20	0	123460	32704	18092	S	0.0	0.4	0:00.16	unattended-upgr
261	root	20	0	5492	2720	2536	S	0.0	0.0	0:00.01	agetty
271	_chrony	20	0	23816	11268	7144	S	0.0	0.1	0:00.10	chronyd
276	_chrony	20	0	12096	2460	1820	S	0.0	0.0	0:00.01	chronyd
327	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
328	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.02	Relay(329)
329	priyada+	20	0	6268	5532	3864	S	0.0	0.1	0:00.09	bash
330	root	20	0	8652	5560	4716	S	0.0	0.1	0:00.01	login
375	priyada+	20	0	22764	13460	10988	S	0.0	0.2	0:00.12	systemd
377	priyada+	20	0	22920	4084	2080	S	0.0	0.1	0:00.00	(sd-pam)